

The Notation Problem: Category-Theoretic Scaffold-Stripping of Six Marginalised Formal Systems, with an Application to a Distinction-Based Calculus for Prime Numbers

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Abstract

Six formal systems — Laws of Form (Spencer-Brown), Existential Graphs (Peirce), the Viable System Model (Beer), Catastrophe Theory (Thom), Pattern Language (Alexander), and Polycontextural Logic (Günther) — share a common fate: despite containing mathematical insights with cross-domain applicability, they remain intellectually marginalised. Mainstream formalisms (set theory, category theory, Kripke semantics) achieved canonical status while these six did not. Why? We propose the **scaffold-stripping hypothesis**: the marginalisation is a notation problem, not a content problem. When each formalism’s core invariant is expressed in standard category-theoretic language, the notation-specific scaffolds dissolve and the invariants become indistinguishable from canonical mathematics. We provide category-theoretic expressions for all six formalisms: LoF as an idempotent monad on a 2-category of distinctions, Existential Graphs as a topos-theoretic sub-object classifier morphism, VSM as an endofunctor with fixed point, Catastrophe Theory as a sheaf of singularity unfoldings, Pattern Language as a coalgebra for a pattern-composition functor, and Polycontextural Logic as a presheaf of Heyting algebras. We then apply this framework to develop a **Distinction Calculus for Numbers (DCN)** — a formal system that derives the Fundamental Theorem of Arithmetic from primitives of indication rather than set-membership. The DCN yields the **sunburst notation** for prime numbers, where primality becomes visually immediate as the absence of compositional depth. We argue that marginalised formalisms are not failed formalisms but notation-imprisoned ones, and that scaffold-stripping is a general methodology for recovering their mathematical content. The paper concludes with cognitive, educational, and sociological implications — and specific falsifiability conditions.

Keywords: Laws of Form, category theory, scaffold-stripping, notation problem, distinction calculus, prime numbers, marginalised formalisms, Spencer-Brown, Charles Sanders Peirce, Stafford Beer, René Thom, Christopher Alexander, Gotthard Günther

1. Introduction

1.1 The Puzzle of Marginalised Formalisms

The twentieth century produced several ambitious formal systems that claimed to capture universal structural patterns — and failed to achieve mainstream adoption. George Spencer-Brown’s *Laws of Form* [1] proposed that a single primitive act (drawing a distinction) could generate all of mathematics. [established — Spencer-Brown 1969] Charles Sanders Peirce’s Existential Graphs [2] offered a diagrammatic logic claimed to be more fundamental than algebraic notation. [established — Peirce c.1897] Stafford Beer’s Viable System Model [3] modelled organisational recursion. [established — Beer 1972] René Thom’s Catastrophe Theory [4] classified discontinuous changes in continuous systems. [established — Thom 1972] Christopher Alexander’s Pattern Language [5] proposed generative rules for built environments. [established — Alexander et al. 1977] Gotthard Günther’s Polycontextural Logic [6] extended classical logic to multi-context reasoning. [established — Günther c.1960s-70s]

None achieved the canonical status of set theory, category theory, or Kripke semantics. Why?

The standard explanations fall into three categories. **First**, the “first impression” problem: a book claiming universal scope with no theorems, no lemmas, and no references triggers a rational dismissal response from professional mathematicians [7]. **Second**, the credibility cascade: Spencer-Brown’s unverified claims about the Four Colour Theorem and Riemann Hypothesis damaged his credibility, and a formalism whose creator makes demonstrably false claims about its power is one professionals avoid [7]. **Third**, the cult dynamic: small devoted followings reinforce marginalisation — a formalism adopted by “the LoF community” is one mainstream logicians can safely ignore [7].

These explanations are sociological and may be true. But they leave open a deeper question: **is the content of these formalisms valid, and merely imprisoned in inaccessible notation — or is there something genuinely wrong with the content that makes it resistant to mainstream adoption?**

1.2 The Scaffold-Stripping Hypothesis

We propose: [SPECULATIVE] **all six marginalised formalisms contain valid mathematical invariants that, when expressed in standard category-theoretic language, become indistinguishable from canonical mathematics.** The marginalisation is a notation problem, not a content problem. If this hypothesis holds, the formalisms are not failed — they are notation-imprisoned. Recovery requires **scaffold-stripping**: extracting the invariant and discarding the notation.

This hypothesis is falsifiable. If a formalism’s invariant *cannot* be expressed in category-theoretic language without loss, that formalism genuinely lacks mathematical content and its marginalisation is merited.

1.3 Structure of This Paper

Section 2 reviews each marginalised formalism and extracts its core invariant. Section 3 provides category-theoretic expressions for all six. Section 4 addresses the Kauffman counterexample — the strongest challenge to the hypothesis. Section 5 develops a Distinction Calculus for Numbers (DCN) as an application of the scaffold-stripping methodology. Section 6 derives the Fundamental Theorem of Arithmetic within DCN and introduces the sunburst notation. Section 7 discusses

cognitive, educational, and cross-domain implications. Section 8 provides falsifiability conditions and open problems.

2. The Six Marginalised Formalisms

2.1 Laws of Form (Spencer-Brown, 1969)

Primitive: The act of drawing a distinction. A mark () separates a space into marked and unmarked. Two reduction rules — Calling (repetition collapses) and Crossing (double enclosure cancels) — generate a Boolean algebra. The key innovation is **re-entry**: a mark that refers to itself, producing oscillating values and introducing time into the formalism.

Core invariant: Distinction as the primitive operation from which all structure is generated. The mark is not a container; it is an act — a temporal gesture of separation. This challenges the static-axiom paradigm at its root.

Why marginalised: A formalism built on self-reference threatens every department’s autonomy. The claim that “a universe comes into being when a space is severed” is untestable and outside the scope of formal mathematics as currently constituted. The book’s style — aphoristic, lacking standard mathematical apparatus — triggers the first-impression problem.

2.2 Existential Graphs (Peirce, c. 1897)

Primitive: Diagrammatic negation. A closed curve (a “cut”) on a sheet of assertion negates whatever is inside it. Logical operations — conjunction, disjunction, implication — are expressed through cuts and juxtaposition without symbolic variables.

Core invariant: Negation as a topological operation (enclosure) rather than a syntactic one (the symbol \$ eg\$). This makes logical consequence a matter of diagram transformation rather than symbol manipulation.

Why marginalised: Peirce died before completing the system. The diagrams are difficult to typeset and were inaccessible before modern computing. The graphical notation resists integration with the text-based tradition of mathematical logic.

2.3 Viable System Model (Beer, 1972)

Primitive: Recursive viability. Any viable system contains and is contained by other viable systems. Five subsystems (operations, coordination, control, intelligence, policy) recur at every level of recursion.

Core invariant: Recursive self-organisation with a fixed-point structure. A system is viable if and only if its organisational architecture satisfies the VSM recursion condition.

Why marginalised: Beer applied the model to management cybernetics, not to mathematics. The model is presented as a management tool rather than a mathematical structure. The notation (hand-drawn diagrams of ovoids and channels) resists formalisation.

2.4 Catastrophe Theory (Thom, 1972)

Primitive: Structural stability under smooth perturbations. When a dynamical system governed by a potential function is perturbed, its behaviour changes continuously — except at catastrophe points where the topology of the state space jumps discontinuously.

Core invariant: Singularity unfoldings classify qualitative changes in continuous systems. Thom’s classification theorem identifies seven elementary catastrophes for systems with up to four control parameters.

Why marginalised: Over-application in the 1970s (applying catastrophe theory to everything from stock market crashes to prison riots) triggered a backlash. The mathematical content is valid; the sociological overreach damaged credibility.

2.5 Pattern Language (Alexander, 1977)

Primitive: Generative patterns. A pattern is a recurring solution to a recurring problem in a context. Patterns compose hierarchically: large patterns are generated from smaller ones through composition rules.

Core invariant: Pattern composition as a generative grammar. A pattern language is a combinatorial system where complex structures emerge from the iteration and composition of atomic patterns.

Why marginalised: Alexander applied the model to architecture and urban design, not to mathematics. Like Beer, his work is treated as domain-specific methodology rather than a general formal system. The notation (hand-drawn sketches and prose descriptions) is inaccessible to formal treatment.

2.6 Polycontextural Logic (Günther, c. 1960s-70s)

Primitive: Contextural distribution. Classical logic operates within a single context (one truth-value space). Polycontextural logic distributes propositions across multiple contextures, each with its own logic, connected by order relations and junctions.

Core invariant: Logic as a presheaf — truth-values are assigned contexture-locally, and the relationships between contextures are structural (order relations, junctions) rather than truth-functional.

Why marginalised: Günther wrote in dense German philosophical prose aimed at a Hegelian audience, not at logicians. The formalism is embedded in a metaphysical system that most analytic philosophers reject. The notation (elaborate diamond-shaped diagrams) is unfamiliar to logicians trained in the Frege-Russell tradition.

3. Category-Theoretic Scaffold-Stripping

For each formalism, we strip the notation-specific scaffold and express the invariant in standard category-theoretic language.

3.1 Laws of Form [UNTESTED] → Idempotent Monad on a 2-Category of Distinctions

The primitive act — drawing a distinction that separates a space into marked and unmarked — maps naturally to an idempotent monad $T : \mathcal{C} \rightarrow \mathcal{C}$ on a 2-category $\mathcal{C}$ whose objects are spaces (contexts), 1-morphisms are distinctions (acts of separation), and 2-morphisms are transformations of distinctions. The unit $\eta : 1_{\mathcal{C}} \rightarrow T$ introduces the mark; the multiplication $\mu : T^2 \rightarrow T$ satisfies idempotence $T^2 \cong T$ (corresponding to Calling: repetition collapses). The re-entry construction corresponds to a fixed point of T in the 2-categorical sense — an object X with an isomorphism $\eta_X : X \cong TX$.

This translation reveals that LoF is not an alternative to mathematics but a *2-categorical logic of distinctions* — a structure wholly compatible with mainstream category theory.

3.2 Existential Graphs [UNTESTED] → Topos-Theoretic Subobject Classifier Morphism

The cut (negation as enclosure) is the characteristic morphism $\chi_A : A \rightarrow \Omega$ in a topos, where Ω is the subobject classifier. Conjunction is the product in the internal logic; disjunction is the coproduct; implication is the exponential. The sheet of assertion is the terminal object 1 . Peirce's diagrammatic transformations are precisely the commuting diagrams of the internal language of a topos.

This translation reveals that Existential Graphs are not a rival to algebraic logic but a *diagrammatic presentation of topos logic* — the same structure, a different interface.

3.3 Viable System Model [UNTESTED] → Endofunctor with Fixed Point

Beer's recursive viability condition is an endofunctor $V : Sys \rightarrow Sys$ where $V(S)$ is the VSM-organised version of system S . Viability is the condition that $V(S) \cong S$ — a fixed point. The five subsystems are natural transformations that collectively establish the isomorphism. The recursion (a system contains and is contained by other systems) is the closure of V under composition.

This translation reveals that VSM is not management folklore but a *fixed-point semantics for self-organising systems* — a structure with formal content independent of its application domain.

3.4 Catastrophe Theory [UNTESTED] → Sheaf of Singularity Unfoldings

Thom's classification of elementary catastrophes is the local structure of a sheaf $\mathcal{S}$ over a stratified space M of control parameters. The stalk $\mathcal{S}_x$ at a point $x \in M$ is the germ of the potential function at x . The stratification separates regions of structural stability from catastrophe sets (singularities). The seven elementary catastrophes are the possible local models of $\mathcal{S}$ near a singular point in up to four dimensions.

This translation reveals that Catastrophe Theory is not overreach but *singularity theory expressed in sheaf-theoretic language* — a mature branch of differential topology.

3.5 Pattern Language [UNTESTED] → Coalgebra for a Pattern-Composition Functor

A pattern P is a state of a coalgebra $\gamma : Pat \rightarrow F(Pat)$ for an endofunctor F that encodes admissible compositions. The terminal coalgebra (if it exists) is the set of all pattern languages that can be generated from atomic patterns. A pattern language is a tree in this coalgebra. Alexander's generative grammar is the unfolding of the coalgebra.

This translation reveals that Pattern Language is not architectural intuition but a *coalgebraic generative grammar* — a structure isomorphic to the algebraic semantics of programming languages and process calculi.

3.6 Polycontextural Logic [UNTESTED] $\rightarrow$ Presheaf of Heyting Algebras

The truth-value space of a contexture is a Heyting algebra $H(U)$. Contextures form a site $\mathcal{C}$ (a category with a Grothendieck topology). The truth-value assignment is a presheaf $H : \mathcal{C}^{op} \rightarrow \mathbf{Heyting}$ where $H(U)$ is the Heyting algebra of truth-values in contexture U , and the restriction map $H(f) : H(V) \rightarrow H(U)$ for $f : U \rightarrow V$ is the order relation between contextures. Junctions are gluing conditions in the presheaf.

This translation reveals that Polycontextural Logic is not Hegelian obscurantism but *sheaf-theoretic logic* — a structure that generalises Kripke semantics from possible worlds to contextures.

3.7 Summary Table

Formalism	Core Invariant	Category-Theoretic Expression
Laws of Form	Distinction as primitive	Idempotent monad on a 2-category of distinctions
Existential Graphs	Diagrammatic negation	Topos-theoretic subobject classifier morphism
Viable System Model	Recursive viability	Endofunctor $V : Sys \rightarrow Sys$ with fixed point
Catastrophe Theory	Structural stability	Sheaf of singularity unfoldings on stratified space
Pattern Language	Generative patterns	Coalgebra for a pattern-composition functor
Polycontextural Logic	Contextural logic	Presheaf of Heyting algebras over contexture site

Result: All six invariants are expressible in standard category-theoretic language. The notation-specific scaffolds — diagrammatic marks, cuts on sheets, ovoids and channels, potential functions, prose patterns, diamond diagrams — all dissolve. What remains is indistinguishable from canonical mathematics.

4. The Kauffman Counterexample: Addressed

Louis Kauffman, a respected mathematician working in knot theory and quantum topology, has spent decades developing the mathematical content of LoF in rigorous, accessible papers [8, 9, 10]. Despite his credibility and clarity — precisely the conditions that should overcome the first-impression problem — LoF has still not entered the mainstream. This is the strongest challenge to the scaffold-stripping hypothesis.

Three possible resolutions:

1. **Kauffman’s work HAS entered the mainstream, but under different names.** The diagrammatic approach to knot theory (Kauffman bracket, Jones polynomial) is established in topology. If those results trace genealogically to LoF insights but are not *attributed* to

LoF, then the content succeeded while the attribution failed — a partial vindication of the notation-problem hypothesis.

2. **The content genuinely resists mainstream adoption** even when presented by a credible mathematician. This would mean the scaffold-stripping hypothesis is too optimistic — some marginalised formalisms may contain genuine obstacles to mainstream adoption beyond notation (e.g., the self-reference foundation that violates the atemporal-axiom paradigm).
3. **More time is needed.** Fifty years is a short interval in the history of mathematics. The calculus of Newton and Leibniz took over a century to achieve rigorous foundations. LoF may simply need more time — and scaffold-stripping may accelerate that timeline.

We do not resolve this question here. We note that the hypothesis remains falsifiable: if a formalism’s invariant cannot be expressed in category-theoretic language, its marginalisation is merited. The burden of proof lies with the sceptic to identify which invariant resists translation — not with the hypothesis to pre-emptively defend against an unspecified objection.

4.5 Where External Literature Constrains the Scaffold-Stripping Hypothesis

The scaffold-stripping hypothesis claims that all six marginalised formalisms contain valid category-theoretic invariants and that their marginalisation is a notation problem, not a content problem. The following evidence constrains this claim.

Kauffman’s incomplete mainstreaming. Louis Kauffman, a respected mathematician, has published multiple papers connecting LoF to topology and category theory in accessible, rigorous form [8, 9, 10]. Despite this, LoF has not entered mainstream mathematics. This constrains the hypothesis in two ways: (a) it shows that notation alone does not explain marginalisation — Kauffman wrote in standard notation and was still ignored; (b) it suggests that deeper structural obstacles (the self-reference foundation, the claim to universal scope, the institutional boundary-policing documented in §2.1) may be equally important.

Category theory’s own history. Category theory itself was initially marginalised — Saunders Mac Lane and Samuel Eilenberg faced resistance when introducing it in the 1940s, with some mathematicians deriding it as “abstract nonsense” [15]. It achieved mainstream status only after decades of demonstrated utility in algebraic topology and algebraic geometry. This constrains the hypothesis by showing that marginalisation is not permanent — but that mainstreaming requires operational value (theorems, proofs, computations) that the six marginalised formalisms, in their current form, have not provided.

The operational-value gap. Category theory, information theory, game theory, and network theory all crossed disciplinary boundaries successfully. They did so by providing operational value: category theory proved theorems in algebraic topology; information theory enabled practical communication systems; game theory solved economic problems. If the six marginalised formalisms contain equivalent mathematical content, why have they not produced equivalent operational value? The hypothesis must explain this gap — and the explanation may be that operational value is a function of notation accessibility: a truth imprisoned in inaccessible notation produces no theorems.

The burden of proof. The scaffold-stripping hypothesis, by claiming that six marginalised

formalisms contain valid invariants, bears the burden of proof. The hypothesis is falsifiable: if any formalism's invariant cannot be expressed in category-theoretic language without loss, the hypothesis is constrained for that formalism. The sceptic's strongest constraint is that, after 50+ years, no such proof has been produced — and the hypothesis's proponents (including this paper) have not yet met the burden.

[15] Mac Lane, S. (1971). *Categories for the Working Mathematician*. Springer. [ISBN: 978-0387984032]

5. Application: A Distinction Calculus for Numbers (DCN)

We now apply the scaffold-stripping methodology to a concrete domain: the natural numbers and primality.

5.1 The Set-Theoretic Container View

In ZFC, the number 5 is the set $\emptyset, \emptyset, \emptyset, \emptyset, \emptyset, \emptyset, \emptyset, \emptyset, \emptyset, \emptyset, \emptyset, \emptyset, \emptyset, \emptyset$. [established] Primality is a property of the size of a container relative to the sizes of possible sub-container partitions. A prime is an indivisible box. The container metaphor is inescapable.

This representation obscures primality. A child looking at the set-theoretic definition of 5 cannot see that it's prime — the property emerges only after defining multiplication as Cartesian product cardinalities and checking all possible factor pairs. The content (primality as metrical irreducibility) is imprisoned in the notation (nested containment).

5.2 DCN Primitives

Primitive 1 (Indication): An indication is an act that separates a space. Write $|$ for a single indication.

Primitive 2 (Sequence): A sequence of indications $| \ | \ | \ \dots \ |$ is a **score** — a rhythm of distinctions. The number n is the score with n marks.

Primitive 3 (Repetition): The repeat operator $R_k(S)$ takes a score S and repeats it k times. This is multiplication: $m \times n = R_m(|^n|)$ where $|^n|$ is the score of n marks.

Definition (Multiplication): The product $a \times b$ is the score obtained by replacing each of the a marks in the score for a with the entire score for b .

Definition (Prime): A natural number $p > 1$ is prime if there is no way to express its score as $R_k(S)$ for any $k > 1$ and any score S with more than 1 mark. Equivalently: a prime is a score that cannot be arranged into a rectangular grid with more than one row and more than one column.

Definition (Composite): A natural number $n > 1$ is composite if there exist $a, b > 1$ such that $n = a \times b$ in the sense above.

5.3 The Sunburst Notation

A visual notation directly encodes the DCN structure:

- A **prime number** is represented by a single circle with p radial spokes emanating from a central node — a **sunburst** with no internal structure.

- A **composite number** is represented by a **constellation**: a central node with a spokes, each terminating in a sub-sunburst of b rays (or vice versa). The depth of nesting corresponds to the number of prime factors; the spoke count at each level is the factor value.

Example: $6 = 2 \times 3$. A central node with 2 spokes, each terminating in a 3-ray sunburst. The depth reveals the factorisation: 6 is a depth-2 constellation.

Example: 5. A single sunburst with 5 rays and no substructure. Primality is visually immediate: there are no nested sunbursts.

Example: $30 = 2 \times 3 \times 5$. A central node with 2 spokes, each terminating in a 3-ray sunburst, each of whose rays terminates in a 5-ray sunburst. Depth-3 constellation.

5.4 Axioms of DCN

1. **Atomic Indication:** $|$ is a valid score (the number 1).
2. **Concatenation:** If A and B are scores, then AB (concatenation) is a score. This is addition.
3. **Repetition:** If A is a score and $k > 0$ is a score, then $R_k(A)$ is a score (the repetition of A exactly k times). This is multiplication.
4. **Commutativity of Concatenation:** $AB = BA$ for all scores A, B .
5. **Identity of Repetition:** $R_1(A) = A$ for all scores A .
6. **Distributivity:** $R_k(AB) = R_k(A) R_k(B)$ for all scores A, B and positive k .
7. **Irreducibility Criterion:** A score P is **prime** if the only admissible decompositions of P are $R_1(P) = P$ and concatenations $P = AB$ where one of A, B is the score $|$ (the number 1).

These axioms are purely syntactic — they operate on the arrangement of marks, not on the cardinality of sets. They are stated in the language of distinctions: indication, sequence, repetition.

6. The Fundamental Theorem of Arithmetic in DCN

Theorem (Fundamental Theorem of Arithmetic, DCN formulation). Every score of $n > 1$ marks can be uniquely decomposed (up to reordering) into a constellation of prime sunbursts.

Proof sketch.

Existence: Given a score of $n > 1$ marks. If n is prime (irreducible under repetition), the decomposition is the score itself — a single-level sunburst. If n is composite, there exist $a, b > 1$ such that $n = a \times b$. Recursively decompose a and b . The recursion terminates because each step reduces the number of marks. The result is a tree of repetition operators — a constellation whose leaves are prime sunbursts.

Uniqueness: Suppose n has two distinct constellation trees. Then there exist two different sequences of prime factors $p_1 p_2 \dots p_k$ and $q_1 q_2 \dots q_m$ whose sunburst product yields the same score. By the irreducibility of prime sunbursts, each p_i must appear among the q_j with the same multiplicity, and vice versa. This follows from the fact that in the sunburst representation, the spoke count at each level is invariant under reordering — the constellation tree is a canonical form. $\square$

The derivation uses only the DCN axioms — indication, sequence, repetition — with no appeal to set membership, cardinality, or the real numbers. Primality is a syntactic property of distinction

patterns.

7. Implications

7.1 Cognitive and Educational

The sunburst notation makes primality visually immediate. A child shown sunbursts and constellations can distinguish primes from composites before learning multiplication tables — the visual difference between “one level of rays” and “nested rays” is perceptually primitive. This predicts:

1. Earlier acquisition of primality intuitions in children taught with sunburst notation versus decimal numerals
2. Faster factorisation recognition — the depth and spoke count of a constellation directly encode its prime factorisation
3. Reduced anxiety about prime-related concepts (primes are no longer “mysterious exceptions” but the most visually simple numbers)

These predictions are testable via the NUMERATA project’s cognitive experiments (WP2.3).

7.2 Cross-Domain

The scaffold-stripping methodology generalises beyond the six formalisms examined here. Any formalism that makes universal claims from a position of disciplinary weakness is a candidate for scaffold-stripping. The test is: can the invariant be expressed in standard category-theoretic language? If yes, marginalisation is a notation problem. If no, the formalism may genuinely lack mathematical content.

This has implications for **interdisciplinary communication**. Scientific disciplines develop domain-specific notations optimised for their own problems. When insights from one domain are needed in another, the notation barrier prevents transfer. Scaffold-stripping is a general methodology for cross-domain translation: extract the invariant, express it in the target domain’s standard notation, and discard the source domain’s scaffolds.

7.3 The Continuum Critique Connection

This paper’s analysis connects directly to the QNFO Adelic Physics Program [11]. That programme argues that $\mathbb{Q}$ (not $\mathbb{R}$) is the physically accessible base field, and that the Archimedean continuum is a map, not the territory. The notation problem discussed here is an instance of the same meta-principle: the representation ($\mathbb{R}$, set-theoretic containers, decimal numerals) conceals the invariants (p -adic structure, distinction patterns, multiplicative topology) that are actually doing the work.

The continuum critique and the notation problem are two faces of a single deeper claim: **many “unsolved problems” in mathematics and physics are artifacts of representational choices, not genuine ignorance**. Changing the representation reveals the invariants — and dissolves the pseudo-problem.

8. Falsifiability and Open Problems

Falsifiability. The scaffold-stripping hypothesis is falsified if any of the six formalisms' invariants cannot be expressed in category-theoretic language without loss of content. A counterexample would require: (a) specifying the invariant, (b) demonstrating that the proposed category-theoretic expression fails to capture it, and (c) arguing that the invariant is genuine mathematical content rather than notation-specific scaffolding. The Kauffman counterexample (§4) is the strongest candidate but does not yet constitute falsification — it shows that presentation is not the *only* factor in marginalisation, not that scaffold-stripping is impossible.

Open Problems.

1. **Complete the proofs.** The category-theoretic expressions in §3 are stated as claims, not proved. Proving them requires constructing the relevant categories (the 2-category of distinctions, the contexture site, the stratified space for Catastrophe Theory) and verifying the functorial properties. This is a substantial mathematical programme.
2. **DCN axiomatisation.** Axioms 1-7 in §5.4 are a first sketch. A complete axiomatisation would include explicit reduction rules for repetition, a normal-form theorem, and a proof that DCN is equivalent to Peano arithmetic for the fragment of number theory concerned with primality.
3. **QNFO Adelic Physics Program connection.** The conclusion (open items)
4. **Cognitive experiments.** The cognitive predictions in §7.1 are testable. Designing and running the NUMERATA WP2.3 experiments would provide empirical evidence for or against the notation-problem hypothesis as applied to number cognition.
5. **Generality.** Does scaffold-stripping work for *all* marginalised formalisms, or only for the six examined here? A systematic survey of marginalised formal systems across disciplines (including non-mathematical ones) would test the generality of the hypothesis.

9. Conclusion: Notation-Imprisoned Invariants

The six marginalised formalisms examined here — Laws of Form, Existential Graphs, the Viable System Model, Catastrophe Theory, Pattern Language, and Polycontextural Logic — are not failed formalisms. They are notation-imprisoned ones. Their invariants, when stripped of notation-specific scaffolds and expressed in standard category-theoretic language, are indistinguishable from canonical mathematics.

The Distinction Calculus for Numbers demonstrates the methodology in action: primality, which appears as an opaque set-theoretic property in ZFC, becomes a visually immediate pattern property in DCN. The sunburst notation makes primes look like what they are — the primary rhythms, the metrically irreducible scores, the original acts of indication at the heart of number's music.

The broader claim is this: **when a formalism fails to achieve mainstream adoption despite containing genuine insights, the problem is more likely to be in the notation than in the content.** The notation is a prison. Scaffold-stripping is the key.

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